Geometrical Configuration of Planes

Questions 1–28

Question 1. Describe all three possible geometrical arrangements of two planes.

Question 2. Describe all eight possible geometrical arrangements of three planes. Ignore replica scenarios in your answer, e.g. $\{1\&2, 3\}$ is the same as $\{1\&3, 2\}$.

Question 3. It is given that Π_1, Π_2 and Π_3 are planes. It is also given that the planes are neither parallel to one another nor coincident. Describe all three possible geometrical arrangements of the three planes.

Question 4. It is given that Π_1, Π_2 and Π_3 are distinct planes. The system of equations representing the planes does not have a unique solution, but the equations are consistent. Describe the geometrical arrangement of the three planes.

Question 5. It is given that $A = \begin{pmatrix} 1 & 4 & -2 \\ 3 & 2 & 2 \\ 5 & -1 & 7 \end{pmatrix}$.

a) Find A^{-1}.

b) Use your answer to (a) to solve the simultaneous equations $x + 4y - 2z = 10$, $3x + 2y + 2z = 12$, $5x - y + 7z = 8$.

c) State the geometrical configuration of the three planes.

Question 6. Three planes have equations $\Pi_1 : 2x + 3y - z = 1$, $\Pi_2 : x + 2y + z = 2$ and $\Pi_3 : 4x - y - z = 3$. Find the unique point of intersection of the three planes.

Question 7. Three planes have equations $\Pi_1 : x+4y-2z = 15$, $\Pi_2 : -6x-7y+z = -40$ and $\Pi_3 : 2x+5y-2z = 22$. Find the unique point of intersection of the three planes.

Question 8. Three planes have equations $\Pi_1 : 4x + y + 9z = k$, $\Pi_2 : 3x + 2y - 5z = 8$ and $\Pi_3 : 4x + 3y - 10z = 3$, where k is a constant. Show that the three planes always meet at a unique point.

Question 9. Determine the geometrical configuration of the following trios of planes:

a) $\Pi_1 : 6x + 2y - 4z = 1$, $\Pi_2 : 5x + y + z = 3$, $\Pi_3 : 3x + y - 2z = 8$

b) $\Pi_1 : 7x - 2y - z = 2$, $\Pi_2 : 9x - 3y + 15z = 12$, $\Pi_3 : 3x - y + 5z = 4$

c) $\Pi_1 : 20x + 30y - 15z = -6$, $\Pi_2 : \frac{4}{3}x + 2y - z = -\frac{2}{5}$, $\Pi_3 : 14x + 21y - \frac{21}{2}z = -\frac{21}{5}$

Question 10. Three planes have equations $\Pi_1 : 11x - 3y + z = 2$, $\Pi_2 : x + 2z = -5$ and $\Pi_3 : 3x - y - z = 4$. Determine their geometrical configuration.

Question 11. Three planes have equations $\Pi_1 : 4x-21y+6z = 10$, $\Pi_2 : 8x+2y+4z = 3$ and $\Pi_3 : 10x+8y+4z = 7$. Determine their geometrical configuration.

Question 12. Three planes have equations $\Pi_1 : 5x+6y+26z = 6$, $\Pi_2 : 2x-3y-4z = 6$ and $\Pi_3 : x-3y-6z = 12$. Determine their geometrical configuration.

Question 13. Three planes have equations $\Pi_1 : 2x + 4y + z = k$, $\Pi_2 : x + 19y - 8z = 0$ and $\Pi_3 : -2x + 4y - 5z = 0$, where k is a constant. Describe their geometrical configuration when (a) $k = 0$, (b) $k \neq 0$.

Question 14. Three planes have equations $\Pi_1 : 2x + ky + kz = k$, $\Pi_2 : kx + 3y + kz = k$ and $\Pi_3 : 4x + 2y + 2z = k$, where k is a constant. It is given that the three planes do not meet at a unique point when $k = 1$ or $k = 3$. Describe their geometrical configuration when (a) $k = 1$, (b) $k = 3$.

Question 15. Three planes have equations $\Pi_1 : 2x+4y+5z = 23$, $\Pi_2 : 3x+ky+4z = 20$ and $\Pi_3 : -2x+4y+kz = 6$, where k is a constant. It is given that the three planes do not meet at a unique point.

a) Find the possible values of k.

b) For each value of k found in (a), identify which configuration of the three planes it represents, fully justifying your answers.

Question 16. Three planes have equations $\Pi_1 : (k+1)x + 3y - 2z = 1$, $\Pi_2 : kx + 4y - \frac{1}{4}(k+8)z = 2$ and $\Pi_3 : 4y - z = 3$, where k is a real constant. Show that the planes always meet at a unique point.

Question 17. Three planes have equations $\Pi_1 : kx - z = 0$, $\Pi_2 : x + 3y + kz = 1$ and $\Pi_3 : x + 2y + kz = 0$, where k is a constant.

a) Show that the planes always meet at a unique point.

b) Find the coordinates of the point of intersection in terms of k.

Question 18. Three planes have equations $\Pi_1 : x+ky-4z = -19$, $\Pi_2 : -6x+4y-3z = -42$ and $\Pi_3 : kx+y+4z = 19$, where k is a constant. It is given that the three planes do not meet at a unique point.

a) Find the possible values of k.

b) For each value of k found in (a), identify which configuration of the three planes it represents, fully justifying your answers.

Question 19. Three planes have equations $\Pi_1 : kx + 2y + 3kz = 2$, $\Pi_2 : 4x + ky - 6z = -2$ and $\Pi_3 : x - 6z = 0$, where k is a constant. It is given that the three planes do not meet at a unique point.

a) Find the possible values of k.

b) For each value of k found in (a), identify which configuration the planes represent, fully justifying your answers.

Question 20. Three planes have equations $\Pi_1 : -5x + 3y + (k+3)z = 2$, $\Pi_2 : kx + 2ky - 4z = -5$ and $\Pi_3 : 2x - 9y + z = 4$, where k is a constant. It is given that the three planes do not meet at a unique point.

a) Find the possible values of k.

b) For each value of k found in (a), identify which configuration the planes represent, fully justifying your answers.

Question 21. Three planes have equations $\Pi_1 : 4x - 2y + 3z = 17$, $\Pi_2 : (k+1)x + 3y + (k-2)z = 15$ and $\Pi_3 : 4x + (k-3)y + 5z = -65$, where k is a constant. It is given that the three planes do not meet at a unique point.

a) Find the possible values of k.

b) For each value of k found in (a), identify which configuration the planes represent, fully justifying your answers.

Question 22. Three planes have equations $\Pi_1 : 2kx + (k+3)y + kz = 3$, $\Pi_2 : x - y + 2z = p$ and $\Pi_3 : 3x + y - 2z = 1$, where k and p are constants. Find the values of k and p such that the planes form a sheaf.

Question 23. The three planes $\Pi_1 : 6x - y + kz = 33$, $\Pi_2 : 2x - 3y + 5z = 25$ and $\Pi_3 : 4x + 2y - 3z = p$, where k and p are constants, form a sheaf.

a) Find the value of k.

b) Find the values of m and n such that $m(6x - y) + n(2x - 3y) \equiv 4x + 2y$ for any values of x and y.

c) By setting $z = 0$ and using your result to (b), find the value of p.

Question 24. It is given that $A = \begin{pmatrix} 3 & -4 & 17 \\ 1 & 10 & 0 \\ 2 & k-1 & 3 \end{pmatrix}$, where k is a constant.

a) Find the values of k for which A^{-1} exists.

For the following plane questions use $\Pi_1 : 3x - 4y + 17z = 17$, $\Pi_2 : x + 10y = 0$ and $\Pi_3 : 2x + 13y + 3z = m$, where m is a constant.

a) Find the unique point of intersection of the three planes in terms of m.

b) For $\Pi_4 : 3x - 4y + 17z = n$, $\Pi_5 : x + 10y = 1$ and $\Pi_6 : 2x + 14y + 3z = 2$, where n is a constant, find n so the planes are consistent. Hence describe their geometrical configuration.

Question 25. Three planes have equations $\Pi_1 : 2x-3y+kz = p$, $\Pi_2 : 5x-3y+k^2z = k-\frac{\sqrt{3}}{2}$ and $\Pi_3 : 4x+12y-z = 8$, where k and p are constants. It is given that the planes do not meet at a unique point.

a) Find the possible exact values of k.

b) Given that the equations are inconsistent, find the possible exact values of p.

c) Hence describe the geometrical configuration of the three planes.

Question 26. It is given that $A = \begin{pmatrix} k-1 & 2-k & 1 \\ k+2 & 1 & -3 \\ k+1 & 2 & -5 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -8 & k+1 \\ 15 & -20 & 15 \\ 7 & -16 & 15 \end{pmatrix}$, where k is a constant.

a) Find AB in terms of k.

b) It is given that $k = 4$. Find AB and hence find A^{-1}.

c) Three planes have equations $\Pi_1 : 3x - 2y + z = 40$, $\Pi_2 : 6x + y - 3z = 100$ and $\Pi_3 : 5x + 2y - 5z = -1000$. Use a matrix method to solve the three equations, and hence describe the geometrical configuration of the three planes.

Question 27. It is given that $A = \begin{pmatrix} 7 & -1 & 2 \\ k & 4 & -2 \\ 1 & -6 & 3 \end{pmatrix}$.

a) Find the value of k such that A is singular.

b) Use a matrix method to solve $7\sqrt{x} - \sqrt{y} + 2\sqrt{z} = 49$, $-\frac{7}{9}\sqrt{x} + 4\sqrt{y} - 2\sqrt{z} = \frac{10}{3}$ and $\sqrt{x} - 6\sqrt{y} + 3\sqrt{z} = -6$.

Question 28. Determine the geometrical configuration of the following trios of planes:

a) $\Pi_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix} + a\begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} + b\begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$; $\Pi_2 : \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix} + m\begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix} + n\begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$; $\Pi_3 : \mathbf{r} = \begin{pmatrix} 8 \\ -3 \\ 2 \end{pmatrix} + p\begin{pmatrix} -2 \\ 4 \\ -3 \end{pmatrix} + q\begin{pmatrix} -1 \\ -1 \\ 3 \end{pmatrix}$.

b) $\Pi_1 : \mathbf{r} = \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix} + a\begin{pmatrix} 1 \\ 5 \\ -2 \end{pmatrix} + b\begin{pmatrix} 3 \\ 9 \\ 5 \end{pmatrix}$; $\Pi_2 : \mathbf{r} = \begin{pmatrix} 3 \\ 10 \\ -10 \end{pmatrix} + m\begin{pmatrix} 6 \\ 18 \\ 10 \end{pmatrix} + n\begin{pmatrix} 3 \\ 15 \\ -6 \end{pmatrix}$; $\Pi_3 : \mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + p\begin{pmatrix} 3 \\ 9 \\ 4 \end{pmatrix} + q\begin{pmatrix} 2 \\ 10 \\ -4 \end{pmatrix}$.